The Thiele Machine

A Complete Mathematical Specification

Derived from the Fully Machine-Checked Coq Corpus

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Abstract

This is the complete mathematical specification of the Thiele Machine: a formal model of computation where structural information carries an explicit, conserved cost μ. Everything here is built from the machine-checked Coq corpus. You can reconstruct the machine from what's in this document alone.

Coq establishes that conclusions follow from axioms. It does *not* establish that axioms model physical reality. That distinction matters and I keep it explicit throughout. Every result is tagged as one of three types: **(S)** structural theorems about the machine as a mathematical object, **(C)** conditional derivations where a physics conclusion follows from a stated axiom, and **(R)** consistency relations that check internal coherence but are not independent predictions.

The document covers: the full state space and operational semantics; No Free Insight; computational universality and Turing strictness; gauge symmetry and Noether conservation; conditional connections to quantum mechanics (Born rule, Tsirelson bound, no-cloning, unitarity, complex amplitudes, Schrödinger equation) under named physical axioms; spacetime emergence and the Landauer bridge; the Thiele Manifold and self-reference tower; and falsifiable predictions with concrete failure modes.

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Chapter 1

Introduction

The Thiele Machine makes the cost of structure explicit. A Turing Machine treats all state transitions as cost-equivalent. The Thiele Machine assigns a specific cost μ to every operation that extracts or asserts structural information. That's the design. Here is where it goes.

The central claim, proven formally: *you cannot gain structural insight about a system without paying μ-cost.* Combined with explicit physical axioms (superposition, linearity, information conservation), that constraint produces structural results that parallel quantum mechanics, thermodynamics, and emergent spacetime. Each result's status — whether it is a structural theorem **(S)**, a conditional derivation **(C)**, or a consistency relation **(R)** — is marked directly. Nothing is hidden.

Eight parts:

1. **Foundations**: State space, instruction set, operational semantics.
2. **Core Theorems**: No Free Insight, μ-Chaitin, cost = complexity.
3. **Computation**: Universality, strict subsumption, confluence, halting.
4. **Symmetry & Conservation**: μ-conservation, Noether/gauge, receipts.
5. **Quantum Mechanics**: Born rule, Tsirelson, no-cloning, unitarity, complex amplitudes, Schrödinger, Planck.
6. **Emergent Structure**: Spacetime metric, causal cones, information causality, Landauer bridge.
7. **Meta-Theory**: Self-reference, Thiele Manifold, Genesis, three-layer isomorphism, Curry–Howard–Thiele.
8. **Predictions**: Falsifiable divergences from standard QM.

1.1 Epistemological Framework

Formal verification (Coq) establishes that conclusions follow logically from axioms. It does **not** establish that axioms model physical reality. This distinction is critical for evaluating the physics claims in Parts V–VIII.

Every result in this specification falls into one of three categories:

(S) Structural Theorems about the Thiele Machine as a mathematical object. These are unconditionally true given the definitions. *Examples*: μ-conservation, gauge invariance, confluence, halting undecidability, strict Turing subsumption.

(C) Conditional Theorems of the form "if axiom X holds, then physics result Y follows." The logical inference is verified; whether axiom X holds in nature is an empirical question. *Examples*: Born rule uniqueness (conditional on linearity), complex necessity (conditional on 2D amplitudes), unitarity (conditional on information conservation).

(R) Consistency Algebraic identities or definitional equivalences that verify internal coherence but do not constitute independent predictions. *Examples*: Planck consistency relation, Tsirelson cost definition.

Each major theorem in Parts V–VIII is annotated with its category. A theorem marked **(C)** or **(R)** is not diminished—conditional derivations and consistency checks are valuable scientific tools—but the reader should not mistake them for derivations from first principles without stated assumptions. See Appendix B for the complete classification table.

Chapter 2

The State Space

2.1 Primitives

Let $\mathbb{N} = \{0, 1, 2, \dots\}$. Let $\mathbb{B} = \{0, 1\}$. Let Σ be a finite alphabet.

2.2 Regions and Modules

Definition 2.1 (Region)**.** A **Region** R is a finite subset of $\mathbb{N}$, canonically represented as a deduplicated list preserving first-occurrence order: $\mathrm{norm}(L) = \mathrm{nodup}(L)$ removes duplicate elements while preserving the order of their first appearance. (The Python VM applies a stricter canonical form via `sorted(set(region))`; the Coq specification uses `nodup Nat.eq_dec` which does not sort.)

Definition 2.2 (Module)**.** A **Module** $M = (R_M, A_M)$ consists of a normalized region $R_M \subset \mathbb{N}$ and a set of axioms $A_M \subset \Sigma^*$.

2.3 The Partition Graph

Definition 2.3 (Partition Graph)**.** A partition graph $G = (N_{id}, \mathcal{T})$ where $N_{id} \in \mathbb{N}$ is the next available module ID and $\mathcal{T} : \{0, \dots, N_{id} - 1\} \rightharpoonup \mathcal{M}$ is a finite partial function from IDs to modules.

Axiom 2.1 (Well-Formedness)**.** *G is well-formed iff $\mathrm{dom}(\mathcal{T}) \subseteq \{0, \dots, N_{id} - 1\}$.*

2.4 Machine State

Definition 2.4 (VM State)**.** The complete state is a 12-tuple:

$$S = (G, C, R, \mathit{Mem}, PC, \mu, T_\mu, \mathit{err}, \ell, m, w, \mathit{cert})$$

where $G \in \mathcal{G}$ is the partition graph, $C \in \mathbb{N}^4$ comprises the CSRs (certificate address, status, error code, heap base), $R \in \mathbb{N}^{32}$ is the register file, $\mathit{Mem} \in \mathbb{N}^{65536}$ is main memory, $PC \in \mathbb{N}$ is the program counter, $\mu \in \mathbb{N}$ is the μ-ledger, $T_\mu \in \mathbb{N}^{16}$ is the flattened 4×4 μ-tensor (row-major), $\mathit{err} \in \mathbb{B}$ is the global error flag (latching), $\ell \in \mathbb{N}$ is the logic-engine accumulator, $m \in \mathbb{N}$ is the machine status register (0 = Turing mode, 1 = Thiele mode), $w \in \mathbb{N}$ is the witness register (`vm_witness`), and $\mathit{cert} \in \mathbb{B}$ is the certified flag (`vm_certified`).

Chapter 3

Instruction Set and Cost Model

3.1 Instruction Categories

The instruction set $\mathcal{I}$ is partitioned into three categories. Every instruction i carries an explicit cost parameter $\Delta\mu_i \in \mathbb{N}$.

3.1.1 Structural Operations

These modify the partition graph G:

- `PNEW`$(R, \Delta\mu)$: Create module with region R.
- `PSPLIT`$(m, L, R, \Delta\mu)$: Split module m into disjoint regions.
- `PMERGE`$(m_1, m_2, \Delta\mu)$: Merge two modules.
- `PDISCOVER`$(m, E, \Delta\mu)$: Attach evidence E (list of axioms) to module m.
- `MDLACC`$(m, \Delta\mu)$: Minimum-description-length accumulate on module m.

3.1.2 Logical Operations

These interact with information content:

- `LASSERT`$(\mathit{freg}, \mathit{creg}, \mathit{kind}, \mathit{flen}, \mathit{cost})$: Assert a formula read from `vm_mem`[`R[freg]`] (encoded-unit count *flen*, eight concrete bits per unit) against a witness package at `vm_mem`[`R[creg]`]. In SAT mode the package contains one satisfying assignment and one falsifying assignment, stored back-to-back; $\Delta\mu = \mathit{flen} \times 8 + S(\mathit{cost}) \geq 1$.
- `LJOIN`$(\mathit{c1reg}, \mathit{c2reg}, \Delta\mu)$: Join certificates from two memory-resident strings addressed by registers.
- `REVEAL`$(m, n, \mathit{cert}, \Delta\mu)$: Reveal n bits of structure from m; cost is $n + S(\Delta\mu)$.
- `EMIT`$(m, p, \Delta\mu)$: Emit payload p from module m; cost is $|p|_{\text{bits}} + S(\Delta\mu)$.

3.1.3 Computational Operations (Reversible ALU)

Reversible register/memory operations:

- `XFER`$(\mathit{dst}, \mathit{src}, \Delta\mu)$: Copy register *src* to *dst*.
- `XOR_LOAD`$(\mathit{dst}, \mathit{addr}, \Delta\mu)$: Load memory word at *addr*.
- `XOR_ADD`$(\mathit{dst}, \mathit{src}, \Delta\mu)$: $\mathit{dst} \leftarrow \mathit{dst} \oplus \mathit{src}$.

- `XOR_SWAP`$(a, b, \Delta\mu)$: Swap registers a and b.
- `XOR_RANK`$(dst, src, \Delta\mu)$: $dst \leftarrow \text{popcount}(src)$.

3.1.4 Standard ALU and Control Flow

- `LOAD_IMM`$(dst, imm, \Delta\mu)$: Load 32-bit immediate into register.
- `LOAD`$(dst, addr, \Delta\mu)$: Load from main memory.
- `STORE`$(addr, src, \Delta\mu)$: Store to main memory.
- `ADD`$(dst, r_1, r_2, \Delta\mu)$: $dst \leftarrow r_1 + r_2 \pmod{2^{32}}$.
- `SUB`$(dst, r_1, r_2, \Delta\mu)$: $dst \leftarrow r_1 - r_2 \pmod{2^{32}}$.
- `JUMP`$(target, \Delta\mu)$: Unconditional jump.
- `JNEZ`$(r, target, \Delta\mu)$: Jump if $R[r] \neq 0$.
- `CALL`$(target, \Delta\mu)$: Push return address, jump to target.
- `RET`$(\Delta\mu)$: Pop return address, jump back.

3.1.5 Auxiliary Operations

- `CHSH_TRIAL`$(x, y, a, b, \Delta\mu)$: Run a CHSH correlation trial; $x, y, a, b \in \{0, 1\}$ required.
- `ORACLE_HALTS`$(payload, \Delta\mu)$: Query halting oracle (requires $\Delta\mu > 0$).
- `CHECKPOINT`$(label, \Delta\mu)$: Execution checkpoint (charges μ).
- `READ_PORT`$(dst, ch, val, bits, \Delta\mu)$: Read from I/O channel (cert-setter).
- `WRITE_PORT`$(ch, src, \Delta\mu)$: Write to I/O channel.
- `HEAP_LOAD`$(dst, addr, \Delta\mu)$: Load from heap-relative address.
- `HEAP_STORE`$(addr, src, \Delta\mu)$: Store to heap-relative address.
- `HALT`$(\Delta\mu)$: Halt execution.

3.1.6 Categorical Morphism Operations

These 7 opcodes (0x27–0x2D) implement the categorical layer over partition modules. See Chapter 28 for laws and proofs.

- `MORPH`$(dst, src, tgt, idx, \Delta\mu)$: Create morphism $src \to tgt$; write morphism ID to `regs`$[dst]$.
- `COMPOSE`$(dst, f, g, \Delta\mu)$: Compose $f; g$ when $f.\text{tgt} = g.\text{src}$; errors on type mismatch.
- `MORPH_ID`$(dst, m, \Delta\mu)$: Create identity morphism id_m (diagonal relation).
- `MORPH_DELETE`$(f, \Delta\mu)$: Remove morphism f; cascades when modules are split/merged.
- `MORPH_ASSERT`$(f, prop, cert, \Delta\mu)$: **Cert-setter.** Charges $S(\Delta\mu) \geq 1$ unconditionally. Even failed attempts (morphism does not exist) incur cost 1. This is the strongest form of the NoFI law: you cannot attempt to certify a relational claim for free.
- `MORPH_TENSOR`$(dst, f, g, \Delta\mu)$: Tensor product $f \otimes g$ (requires union modules to exist).

- `MORPH_GET`$(dst, f, sel, \Delta\mu)$: Read morphism metadata: selector $0 =$ src, $1 =$ tgt, $2 =$ coupling length, $3 =$ is_identity.

The active implementation exposes a 46-opcode ISA (39 original + 7 categorical morphism opcodes). This condensed mathematical specification presents the core instruction families relevant to the kernel semantics and cost model. Every instruction carries an explicit $\Delta\mu$ parameter; the function `instruction_cost` extracts it.

3.2 Cost Model

Every instruction i carries an explicit cost parameter $\Delta\mu_i \in \mathbb{N}$. The function $\texttt{instruction\_cost}(i)$ extracts this parameter. The cost is applied via $\texttt{apply\_cost}(s, i) = s.\mu + \texttt{instruction\_cost}(i)$. Since $\Delta\mu_i : \mathbb{N}$, the μ-ledger is monotonically non-decreasing by construction.

Semantically, only `REVEAL` and `LASSERT` (when adding structure) constitute genuine information-theoretic cost. Other instructions may carry $\Delta\mu > 0$ as a bookkeeping charge, but in typical programs their cost is set to zero.

Chapter 4

Operational Semantics

4.1 Transition Function

The transition $\delta : \mathcal{S} \times \mathcal{I} \to \mathcal{S}$ is defined by:

1. $PC' = PC + 1$.
2. $\mu' = \mu + \Delta\mu_i$.
3. $err' = err \vee \mathrm{fail}(S, i)$.

4.2 Traces and Execution

Definition 4.1 (Trace)**.** A trace $\tau = [i_0, \ldots, i_k] \in \mathcal{I}^*$ is a finite sequence of instructions.

Definition 4.2 (Execution)**.** $\mathrm{Run}([], S) = S; \quad \mathrm{Run}(i :: \tau', S) = \mathrm{Run}(\tau', \delta(S, i))$.

4.3 Selected Instruction Semantics

PNEW: Normalize region; if already present, idempotent; else add $(R_{\mathrm{norm}}, \emptyset)$ to G.

REVEAL: Update CSRs with checksum of certificate *cert*. Cost: $bits + S(\Delta\mu)$.

LASSERT: Read formula from `vm_mem`[`R[formula_addr_reg]`] (encoded-unit count `formula_len`, eight concrete bits per unit) and witness package from `vm_mem`[`R[cert_addr_reg]`]. Verify using `cert_kind` (false = LRAT/UNSAT, true = satisfying model plus falsifying countermodel). If valid, append the formula to A_m; if invalid, set $err \leftarrow$ true. Cost: `formula_len` $\times\, 8 +$ $S($`mu_delta`$)$.

Chapter 5

No Free Insight

The central theorem: structural insight is never free.

5.1 Definitions

Definition 5.1 (Receipt Predicate)**.** $P : \mathcal{O} \to \mathbb{B}$ is a computable function on observation histories.

Definition 5.2 (Strength Ordering)**.** $P_1 \leq P_2$ iff $\forall o.\ P_1(o) = 1 \implies P_2(o) = 1$. $P_1 < P_2$ iff $P_1 \leq P_2$ and $\exists o.\ P_2(o) = 1 \wedge P_1(o) = 0$.

Definition 5.3 (Certification)**.** Trace τ **certifies** P iff execution succeeds ($err = 0$), the supra-certificate flag is set, and the output satisfies P.

Definition 5.4 (Structure Addition)**.** HasStructureAddition(τ) is true iff τ contains a `REVEAL`, `LASSERT`, or equivalent operation.

5.2 The Theorem

Theorem 5.1 (No Free Insight (NoFreeInsight_Theorem.v))**.** *Let $P_{\text{strong}} < P_{\text{weak}}$. If a trace τ starting from a clean state s_0 certifies P_{strong}, then:*

$$\text{Certified}(\tau, P_{\text{strong}}) \implies \text{HasStructureAddition}(\tau)$$

Proof Sketch. The proof is parametric: `NoFreeInsight_Theorem.v` proves the theorem for any system satisfying the `NO_FREE_INSIGHT_SYSTEM` module type interface. The interface requires four contract obligations (not Coq `Axiom` declarations—they are proved separately for the kernel instantiation): (1) μ-monotonicity, (2) a certification specification linking `certifies` to execution, (3) a no-free-insight contract stating that strict strengthening requires structure addition, and (4) that the system has a notion of clean start. The theorem file itself contains zero `Axiom`, zero `Admitted`. The contract obligations are discharged by the kernel instantiation. □

Corollary 5.2 (Cost of Insight)**.** *Gaining insight (strengthening the predicate) implies $\Delta\mu > 0$.*

Chapter 6

The μ-Chaitin Theorem

A Chaitin-style incompleteness theorem denominated in μ-currency.

Theorem 6.1 (μ-Chaitin (MuChaitinTheory_Theorem.v))**.** *For any formal theory system T satisfying the μ-Chaitin interface, the number of bits k the theory can certify is bounded by:*

$$k \leq |T| + c$$

where $|T|$ is the description length of the theory and c is a fixed overhead constant. The proof chain: $k \leq \text{payload} \leq \mu\text{-}\textit{info} \leq \text{budget} \leq |T| + c$.

Chapter 7

Cost Equals Kolmogorov Complexity

Theorem 7.1 (μ-Bits = Prefix-Free Complexity (CostIsComplexity.v — reference-only))**.** *Let* $K(\text{spec})$ *denote the prefix-free Kolmogorov complexity of specification* spec*. Then:*

$$\mu(\text{spec}) = K(\text{spec})$$

Specifically, $\mu(\text{spec}) = |\text{spec}| + 1$ *(the terminating bit), which equals the minimum description length over all prefix-free programs producing* spec*.*

Proof. The canonical compiler maps $\text{spec} \mapsto \text{spec} \| [\text{true}]$. Any producing program has the form $p = \text{spec} \| [\text{true}]$ with $|p| = |\text{spec}| + 1$. The compiler achieves the minimum, and μ counts exactly these bits. $\square$

Chapter 8

No Free Lunch and No Arbitrage

8.1 No Free Lunch (Ghosts Are Impossible)

Theorem 8.1 (No Free Lunch (NoFreeLunch.v — reference-only)). *A "ghost" is defined as two distinct propositions $p \neq q$ represented by the same physical state. Ghosts are impossible:*

$$\forall\, p \neq q,\ \nexists\, s \text{ s.t. } s \text{ faithfully represents both } p \text{ and } q$$

Information cannot exist without physical distinction.

8.2 No Arbitrage Implies Potential Function

Theorem 8.2 (Potential from No-Arbitrage (NoArbitrage.v — reference-only)). *Let $w : \mathcal{I}^* \to \mathbb{Z}$ be a cost function satisfying:*

1. ***Additivity**: $w([]) = 0$ and $w(\tau_1 \cdot \tau_2) = w(\tau_1) + w(\tau_2)$.*
2. ***No-Arbitrage**: For every closed cycle τ (returning to the starting state), $w(\tau) \geq 0$.*

Then there exists a potential function $\phi : \mathcal{S} \to \mathbb{Z}$ such that:

$$w(\tau) \geq \phi(\text{apply}(\tau, s)) - \phi(s)$$

for every state s and trace τ.

Remark 8.1. This is structurally analogous to the Second Law of Thermodynamics: consistent cost accounting implies a potential function bounding all transitions. The concrete Coq model is a toy (increment/decrement on nat with asymmetric costs). No temperature, entropy, or thermodynamic system appears in the formalization. The connection to thermodynamics is an analogy, not a derivation.

Chapter 9

Computational Universality and Subsumption

9.1 Turing Machine Embedding

Theorem 9.1 (Abstract Simulation (Embedding_TM.v — reference-only, modular_proofs/TM_to_Minsky.v — reference-only))**.** *For every Turing Machine T, there exists a Minsky counter machine M and a simulation relation R such that M simulates T step-by-step, encoding the tape into two integers:*

$$\text{Left} = \sum_{i=0}^{h-1} t[h-1-i]\, 2^i, \qquad \text{Right} = \sum_{i=0}^{n-h-1} t[h+i]\, 2^i$$

The Thiele Machine simulates Minsky using its reversible ALU. Hence:

$$\text{TM} \preceq \text{Minsky} \preceq \text{Thiele}$$

9.2 Strict Containment

Theorem 9.2 (Turing $\subsetneq$ Thiele (kernel/Subsumption.v))**.** *Classical Turing computation is* ***properly extended*** *by sighted Thiele computation (both are Turing-complete, but Thiele carries additional partition and μ-accounting structure):*

$$\exists\, p.\ \text{sighted}(p) \land \neg\text{turing}(p)$$

There exist programs that are Thiele-computable (using partition structure and μ-accounting) but are not classical Turing programs. The extra structure is genuine new content.

Theorem 9.3 (Semantic Strictness (ThieleFoundations.v — reference-only))**.** *There exist Thiele traces with identical Turing shadows but non-isomorphic Thiele-level behavior. The partition and μ layers carry information not present in the Turing skeleton.*

9.3 Halting Undecidability

Theorem 9.4 (Diagonal Argument (kernel/OracleImpossibility.v — reference-only))**.** *No total computable function decides the halting problem for the Thiele Machine.*

9.4 Oracle Cost Lower Bound

Theorem 9.5 (Oracle Impossibility (kernel/OracleImpossibility.v — reference-only))**.** *The formalization defines a* sound *oracle as one charging $\Delta\mu \geq 1$ per query (`sound_oracle_cost`). Given this definition:*

1. *`oracle_halts_costs_mu`: A zero-cost oracle violates the soundness definition ($0 \geq 1 \Rightarrow \perp$).*

2. *`oracle_cost_linear`: Given `sound_oracle_cost`, n queries cost $\geq n$.*

These results verify the internal consistency *of the pricing definition. The physical justification for why oracle queries should cost $\mu > 0$ comes from the No Free Insight framework (gaining information about undecidable propositions should cost something), but the Coq proofs verify the narrower claim that the pricing is self-consistent.*

9.5 Confluence

Theorem 9.6 (Church–Rosser (Confluence.v — reference-only))**.** *For any state s and two independent certificates c_1, c_2 (targeting distinct CNF formulas), applying them in either order yields the same μ-cost:*

$$\mu(s_{12}) = \mu(s_{21})$$

Independent structural operations commute.

9.6 Tensoriality

Theorem 9.7 (Module Independence (ThieleUnificationTensor.v — reference-only))**.** *For distinct modules $m_1 \neq m_2$, recording discoveries commutes:*

$$\text{record}(\text{record}(G, m_1, e_1), m_2, e_2) \equiv_{\text{ext}} \text{record}(\text{record}(G, m_2, e_2), m_1, e_1)$$

Local per-module operations are tensorial.

9.7 Amortized Analysis

Theorem 9.8 (Amortization (AmortizedAnalysis.v — reference-only))**.** *For a batch of T instances with discovery cost D and operational cost O per instance:*

$$\text{total\_cost} = B \cdot D + T \cdot O \geq T \cdot O$$

where B is batch count. As $T \to \infty$, average cost $\to O$ (discovery overhead amortizes away).

Chapter 10

μ-Conservation and Receipt Integrity

10.1 The Conservation Law

Theorem 10.1 (μ-Ledger Conservation (MuLedgerConservation.v)). *For any trace $\tau = [i_0, \ldots, i_k]$:*

$$\mu_{\text{final}} = \mu_{\text{init}} + \sum_{j=0}^{k} \text{cost}(i_j)$$

Every instruction increases μ by exactly its declared cost. No axioms, no admits.

Corollary 10.2 (Monotonicity). *μ never decreases: $\mu(s) \leq \mu(\text{Run}(\tau, s))$ for all τ.*

Corollary 10.3 (Irreversibility Bound). $\text{irreversible\_bits}(\tau) \leq \mu_{\text{final}} - \mu_{\text{init}}$.

Theorem 10.4 (μ Decomposition (MuLedgerConservation.v)). *$\mu_{\text{total}} = \mu_{\text{blind}} + \mu_{\text{sighted}}$ for any partition of the cost into blind (reversible) and sighted (structural) components.*

10.2 Receipt Integrity

Definition 10.1 (Receipt). A receipt $r = (\text{step}, \text{instr}, \mu_{\text{pre}}, \mu_{\text{post}}, h_{\text{pre}}, h_{\text{post}})$ records one transition.

Theorem 10.5 (Receipt Integrity (ReceiptIntegrity.v)). *A valid receipt or execution attestation proves $\mu_{\text{final}} = \mu_{\text{init}} + \sum \text{costs}$ for the attested run. Any receipt claiming $\Delta\mu \neq \text{cost}(\text{instr})$ fails validation.*

Theorem 10.6 (Non-Forgeability). *Forged receipts (claiming incorrect $\Delta\mu$) fail validation. Overflow values ($\mu > 2^{31} - 1$) are rejected.*

Chapter 11

Gauge Symmetry and Noether's Theorem

11.1 The $\mathbb{Z}$-Action (Gauge Group)

Definition 11.1 (Gauge Shift)**.** For $\delta \in \mathbb{Z}$, the gauge shift $\sigma_\delta : \mathcal{S} \to \mathcal{S}$ maps $S \mapsto S[\mu \leftarrow \mu + \delta]$, preserving all other fields.

Theorem 11.1 (Group Action (KernelNoether.v))**.** *The gauge shifts form a $\mathbb{Z}$-action on states:*

1. ***Identity**: $\sigma_0(S) = S$.*
2. ***Composition**: $\sigma_a(\sigma_b(S)) = \sigma_{a+b}(S)$.*
3. ***Inverse**: $\sigma_{-n}(\sigma_n(S)) = S$ (when $\mu + n \geq 0$).*

11.2 Gauge Invariance

Definition 11.2 (Observable)**.** Obs(S) extracts the partition region structure, ignoring μ.

Theorem 11.2 (Gauge Invariance (KernelNoether.v))**.** Obs($\sigma_\delta(S)$) = Obs(S). *Shifting μ by a constant does not affect observables.*

Theorem 11.3 (Noether's Theorem (KernelNoether.v, RepresentationTheorem.v — reference-only))**.**

1. ***Forward**: States identical except for μ lie on the same gauge orbit. (This is direct from the definition: if all other fields match, the μ difference is the gauge shift.)*
2. ***Gauge Invariance**: `z_gauge_invariance` proves that shifting μ does not change `Observable_partition`. This is definitional: `z_gauge_shift` only modifies the `vm_mu` field, and `Observable_partition` reads only `vm_graph`.*
3. ***Trace Preservation**: Gauge-equivalent states produce identical observable traces (same labels, same μ-costs) for any finite horizon. Proven as `vm_step_orbit_equiv`.*

The μ-ledger is a gauge degree of freedom. Its absolute *value is unobservable; only* differences *$\Delta\mu$ are physical. The "Noether" label is a deliberate analogy—the result follows from the record structure (separate fields for μ and graph), not from a deep symmetry principle.*

Theorem 11.4 (μ-Monotonicity (KernelNoether.v))**.** vm_step$(S, i, S') \implies \mu(S) \leq \mu(S')$. *The μ-ledger never decreases under the dynamics. This is true by construction: instruction costs are natural numbers (≥ 0), so adding them to the ledger cannot decrease it. The proof examines each step constructor and confirms the arithmetic.*

Chapter 12

Two-Dimensional Amplitude Space

Remark 12.1 (Epistemological Status: **(C)** Conditional Derivation)**.** The following derivation assumes that partition states admit *amplitude* representations (superpositions), not merely classical probability distributions. This is stated as Axiom 12.1. Without it, classical probability theory ($p \in [0, 1]$, one-dimensional, continuous) suffices and the 2D argument does not apply. The passage from a discrete binary partition to the continuum S^1 also requires a limiting process not formalized in the Coq suite. The argument does not rule out quaternionic (4D) amplitudes; it establishes 2D as the *minimum* dimension for continuous superposition.

Axiom 12.1 (Superposition Principle)**.** Partition module states admit amplitude representations: a state with n classical configurations is described by n real amplitudes $(a_1, \ldots, a_n)$ satisfying $\sum_i a_i^2 = 1$, where a_i^2 gives the probability of configuration i.

Given this axiom, binary partition structure forces quantum amplitudes to live in at least two dimensions.

Theorem 12.1 (1D Is Insufficient (TwoDimensionalNecessity.v — reference-only))**.** *A one-dimensional normalized amplitude satisfies $x^2 = 1$, hence $x \in \{+1, -1\}$. No intermediate superpositions exist.*

Theorem 12.2 (2D Is Necessary and Sufficient (TwoDimensionalNecessity.v — reference-only))**.** *Binary partition structure requires exactly 2D amplitude space. Two-dimensional amplitudes (a, b) with $a^2 + b^2 = 1$ admit a continuous family via $a = \cos\theta$, $b = \sin\theta$: the unit circle S^1.*

Chapter 13

Complex Amplitudes from Norm Preservation

Zero-cost evolution must preserve the norm $a^2 + b^2 = 1$. The group of norm-preserving maps on S^1 is $\mathrm{SO}(2) \cong U(1)$, forcing amplitudes to be complex numbers.

Theorem 13.1 (Rotation Group (ComplexNecessity.v — reference-only))**.** *2D rotations* $R_\theta(a, b) = (a\cos\theta - b\sin\theta,\ a\sin\theta + b\cos\theta)$ *satisfy:*

1. $|R_\theta(a, b)|^2 = a^2 + b^2$ *(norm preservation).*
2. $R_0 = \mathrm{id}$ *(identity).*
3. $R_{\theta_1} \circ R_{\theta_2} = R_{\theta_1+\theta_2}$ *(composition = addition).*
4. $R_\theta^{-1} = R_{-\theta}$ *(invertibility).*

Theorem 13.2 (Complex Necessity (ComplexNecessity.v — reference-only))**.** *Complex multiplication by* $e^{i\theta}$ *is exactly 2D rotation. The norm-preserving maps on* S^1 *are exactly complex multiplications:* $\mathrm{SO}(2) \cong U(1)$*. Hence quantum amplitudes* must *be complex numbers.*

Corollary 13.3 (Euler's Formula)**.** $e^{i\theta_1} \cdot e^{i\theta_2} = e^{i(\theta_1+\theta_2)}$.

Derivation chain: Binary partition $\to$ 2D amplitudes $\to$ normalization (S^1) $\to$ zero-cost $\Rightarrow$ norm-preserving $\Rightarrow$ $\mathrm{SO}(2) \cong U(1)$ $\Rightarrow$ complex numbers.

Chapter 14

The Born Rule

Definition 14.1 (Bloch Sphere Probabilities). For a state with purity vector (x, y, z):

$$P(|0\rangle) = \frac{1+z}{2}, \qquad P(|1\rangle) = \frac{1-z}{2}$$

Definition 14.2 (Measurement μ-Cost).

$$\text{Cost}(x, y, z) = \frac{1 - (x^2 + y^2 + z^2)}{2}$$

This is the linear entropy. For pure states ($r^2 = 1$), cost $= 0$. For mixed states ($r^2 < 1$), cost > 0.

Theorem 14.1 (Uniqueness of the Born Rule (BornRule.v)). *The Born rule $P(|0\rangle) = (1 + z)/2$ is the **unique** probability assignment satisfying:*

1. *Non-negativity and normalization: $P \geq 0$, $P(|0\rangle) + P(|1\rangle) = 1$.*
2. *Eigenstate consistency: $P(0, 0, 1) = 1$, $P(0, 0, -1) = 0$.*
3. *Linearity in z: $P(x, y, z, 0) = az + b$ for some a, b.*

Proof: Boundary conditions force $a = 1/2$, $b = 1/2$. No other solution exists.

Note: *The Coq theorem also lists a `mu_consistent` hypothesis, but it is **unused** in the proof. The hypothesis `mu_consistent P` unfolds to "measurement cost ≥ 0 for valid Bloch vectors," which is directly from the definition true for all states and does not constrain P. The linearity assumption (item 3) does all the work—it assumes the functional form $P = az + b$ and solves a 2-equation system. This is less "deriving the Born rule from μ-accounting" and more "the Born rule is the unique linear rule consistent with eigenstates." The non-circular derivation from tensor consistency (`BornRuleFromSymmetry.v`, below) is the stronger result.*

Theorem 14.2 (Born Rule from Tensor Consistency (BornRuleFromSymmetry.v — reference-only/replaced; active derivation in `kernel/BornRuleLinearity.v`)). *Let $g : [0, 1] \to [0, 1]$ be any function satisfying:*

1. *Eigenstate consistency: $g(0) = 0$, $g(1) = 1$.*
2. *Normalization: $g(x) + g(1 - x) = 1$ for all $x \in [0, 1]$.*
3. *Tensor product consistency: $g(xy) = g(x) \cdot g(y)$ for $x, y \in [0, 1]$.*
4. *Range: $0 \leq g(x) \leq 1$ for $x \in [0, 1]$.*

Then $g(x) = x$ for all $x \in [0, 1]$.

Remark 14.1 (Epistemological Status: **(C)** Conditional Derivation—Circularity Resolved)**.** The original proof (`BornRule.v`) depends on the linearity axiom (item 3): P is affine in z. This assumption is equivalent to the Born rule restated, raising a circularity concern.

The new derivation (`BornRuleFromSymmetry.v`) replaces linearity with *tensor product consistency*: independent measurements yield independent outcomes, i.e., $g(xy) = g(x)g(y)$. This is a structural property of the machine, not a physics axiom—it follows from module independence proven in `ThieleUnificationTensor.v`. The proof proceeds: $g = \mathrm{id}$ on all dyadic rationals (by multiplicativity + normalization); then monotonicity + density of dyadics forces $g = \mathrm{id}$ everywhere (via the Archimedean property of $\mathbb{R}$). Zero admits.

Chapter 15

Unitarity

Definition 15.1 (Evolution)**.** An evolution E maps Bloch vectors $(x, y, z) \mapsto (x', y', z')$ with associated μ-cost.

Theorem 15.1 (Unitarity from Zero Cost (Unitarity.v))**.**

1. $\mu = 0 \implies r'^2 \geq r^2$ *(purity cannot decrease at zero cost). Proven as* `zero_cost_preserves_purity`*.*
2. *Non-unitary evolution ($r'^2 < r^2$ for some state) requires $\mu > 0$. Proven as* `nonunitary_requires_mu`*: the proof specializes the conservation hypothesis at the witness, then* `lra`*.*
3. *$\mu = 0 \Rightarrow$ no info loss (one direction proven). The converse and the full equivalence $\mu = 0 \Leftrightarrow$ unitary $\Leftrightarrow$ reversible are* stated *in the file but not proven as theorems—they appear as* `Definition` *declarations (propositional constants) rather than proven* `Theorem`*s.*

Theorem 15.2 (Lindblad Requires μ (Unitarity.v))**.** *Lindblad-type dissipation at rate $\gamma > 0$ requires $\mu \geq \gamma$. Decoherence is paid for.*

Theorem 15.3 (CPTP Structure (Unitarity.v))**.** *Positivity + trace preservation $\implies$ the evolution is CPTP (completely positive, trace-preserving).*

Chapter 16

Purification

Theorem 16.1 (Purification Principle (Purification.v)). *Every mixed state* (x, y, z) *with* $r^2 = x^2 + y^2 + z^2 \leq 1$ *admits eigenvalues* λ_1, λ_2 *with:*

- $\lambda_1 + \lambda_2 = 1, \quad 0 \leq \lambda_i \leq 1.$
- $(\lambda_1 - \lambda_2)^2 = r^2.$
- *Construction:* $\lambda_1 = (1 + r)/2$, $\lambda_2 = (1 - r)/2$.

Pure states ($r = 1$) *need no external reference (deficit* $= 0$*). The maximally mixed state* ($r = 0$) *has deficit* $= 1$.

Chapter 17

No-Cloning

Theorem 17.1 (No-Cloning from μ-Conservation (NoCloning.v))**.** *Let $I = x^2 + y^2 + z^2$ be the information content (purity). For a cloning operation with outputs of information I_1, I_2 and μ-cost μ:*

$$I_1 + I_2 \leq I + \mu$$

Theorem 17.2 (No-Cloning from μ-Monotonicity (NoCloningFromMuMonotonicity.v — reference-only/replaced; active proof in `kernel/NoCloning.v`))**.** *Machine-native no-cloning using nat-valued structural content (MDL complexity). If cloning duplicates n bits of structural content while producing zero new bits: $2n \leq n + 0$ implies $n \leq 0$. The entire proof is a single `lia` step (linear integer arithmetic). No Bloch sphere, no Hilbert space, no real analysis. Zero admits.*

Corollary 17.3 (Perfect Cloning Is Impossible at Zero Cost)**.** *$I_1 = I_2 = I$ and $\mu = 0$ implies $2I \leq I$, hence $I \leq 0$. Only the trivial state can be cloned for free.*

Corollary 17.4 (Cloning Cost)**.** *Perfect cloning requires $\mu \geq I$. For pure states: $\mu \geq 1$.*

Theorem 17.5 (Approximate Cloning (NoCloning.v))**.** *For fidelities f_1, f_2: $f_1 + f_2 \leq 1 + \mu/I$. At zero cost: $f_1 + f_2 \leq 1$.*

Theorem 17.6 (No Deletion (NoCloning.v))**.** *Perfect deletion also requires $\mu \geq I$ (dual of no-cloning).*

Chapter 18

Observation and Collapse

18.1 Observation Irreversibility

Theorem 18.1 (REVEAL Is Irreversible (ObservationIrreversibility.v — reference-only))**.** *For* bits > 0*:* $\mu_{\text{after}} > \mu_{\text{before}}$*, and the post-REVEAL state* $\neq$ *the pre-REVEAL state. Observation prevents recovery of the pre-measurement superposition.*

18.2 Collapse Determination

Definition 18.1 (Partition Entropy)**.** $H(P) = \log_2(\dim(P))$ where dim is the partition state dimension.

Theorem 18.2 (Maximum Information Implies Unique State (CollapseDetermination.v — reference-only))**.** *If bits revealed* $= H(P_{\text{before}})$ *(maximum information), then* $\dim(P_{\text{after}}) = 1$*. The measurement fully collapses the state to a unique outcome. This is the* ***projection postulate****, derived rather than assumed.*

Chapter 19

The Tsirelson Bound

Definition 19.1 (Correlation μ-Cost)**.** $\mu_{\text{corr}}(S) = 0$ if $|S| \le 2\sqrt{2}$; otherwise $\mu_{\text{corr}} = |S| - 2\sqrt{2}$.

Remark 19.1 (Epistemological Status: **(C)** Conditional Derivation—Circularity Resolved)**.** The original Coq proof (`TsirelsonDerivation.v` — reference-only/replaced) encoded $2\sqrt{2}$ in the definition of μ_{corr}, making its derivation status **(R)**.

This gap is now closed. `TsirelsonGeneral.v` derives $S^2 \le 8$ from pure algebra: a sum-of-squares identity proves each CHSH row contributes $\le 2(e_1^2 + e_2^2)$, then Cauchy–Schwarz bounds the sum by $4\sum e_i^2 \le 8$. `TsirelsonFromAlgebra.v` provides a self-contained algebraic derivation with achievability witness ($e = \pm 1/\sqrt{2}$, giving $S = 2\sqrt{2}$) and a verified rational bound ($\sqrt{8} < 5657/2000$). Zero admits in both files.

Theorem 19.1 (Tsirelson from Zero μ (TsirelsonDerivation.v — reference-only/replaced) — **(R)**)**.** *Total $\mu = 0$ implies $|S_{\text{CHSH}}| \le 2\sqrt{2} \approx 2.828$.*

Theorem 19.2 (CHSH Separation (Deliverable_CHSHSeparation.v — reference-only))**.** *Strict numerical separation:*

$$2 \quad < \quad 2\sqrt{2} \approx 2.828 \quad < \quad \frac{16}{5} = 3.2$$

- ***Classical*** *(local receipts): $|S| \le 2$ (Bell bound).*
- ***Quantum*** *(admissible, $\mu = 0$): $|S| \le 2\sqrt{2}$ (Tsirelson bound).*
- ***Supra-quantum*** *($\mu > 0$ allowed): witnesses achieve $|S| = 3.2$.*

The Tsirelson bound $2\sqrt{2}$ is the maximum correlation purchasable at zero μ-cost. Exceeding it requires paying for additional structural information.

Chapter 20

The Schrödinger Equation

Theorem 20.1 (Emergent Schrödinger (EmergentSchrodinger.v — reference-only))**.** *The finite-difference Schrödinger equation emerges from the partition update rules. For a two-component amplitude state* (a, b) *with mass* m*, potential* V*, and time step* Δt*:*

$$a(t + \Delta t) = a - \frac{\Delta t}{2m}\nabla^2 b + \Delta t \cdot Vb \tag{20.1}$$

$$b(t + \Delta t) = b + \frac{\Delta t}{2m}\nabla^2 a - \Delta t \cdot Va \tag{20.2}$$

The coupling is ***antisymmetric****:* $c_{a,\nabla^2 b} = -c_{b,\nabla^2 a}$ *and* $c_{a,Vb} = -c_{b,Va}$*, which is necessary and sufficient for probability conservation* $\partial_t(a^2 + b^2) = 0$*.*

Remark 20.1 (Epistemological Status: **(C)** Conditional Derivation)**.** This is exactly the finite-difference discretization of $i\hbar\partial_t\psi = -\frac{\hbar^2}{2m}\nabla^2\psi + V\psi$ with $\psi = a + ib$. The Schrödinger form emerges *conditionally*: given Axiom 12.1 (which entails 2D amplitudes) and the requirement that zero-cost evolution preserves probability ($\partial_t(a^2 + b^2) = 0$), the antisymmetric coupling structure is the unique linear solution. The mass m and potential V are parameters, not derived.

Chapter 21

Planck's Constant

Remark 21.1 (Epistemological Status: **(R)** Consistency Relation)**.** The following is **not** a derivation of Planck's constant from first principles. It is a consistency relation that gives h a physical interpretation within the μ-framework. The Coq proof verifies an algebraic identity; the physics is in the definitions.

Definition 21.1 (Physical Constants (Normalized Units))**.** $k_B = 1/100, \quad T = 1, \quad h = 1.$ Landauer energy: $E_L := k_B T \ln 2$.

Definition 21.2 (Computational Time Step)**.** $\tau_\mu := h/(4E_L)$, the Margolus–Levitin time at Landauer energy.

Theorem 21.1 (Planck Consistency Relation (PlanckDerivation.v — reference-only) — **(R)**)**.**

$$h = 4 \cdot E_L \cdot \tau_\mu$$

Remark 21.2 (Why This Is Circular)**.** Substituting $\tau_\mu = h/(4E_L)$ yields $h = 4E_L \cdot h/(4E_L) = h$. The Coq proof reduces to the `field` tactic (pure algebra). This does *not* predict h; it defines τ_μ in terms of h and verifies consistency.

What the relation does provide: a physical interpretation of Planck's constant as $4\times$ the action (energy $\times$ time) of one μ-bit operation at Landauer cost. If taken as a physical claim, the testable content is the predicted time step $\tau_\mu = h/(4k_B T \ln 2)$, which at room temperature yields $\tau_\mu \approx 5.7 \times 10^{-14}$ s (femtosecond range, consistent with molecular vibration timescales).

What would make this non-circular: defining τ_μ as an independent primitive (measured experimentally) and then *deriving* $h = 4E_L\tau_\mu$ as a *prediction*. The current formalization does not do this.

Chapter 22

Emergent Spacetime

22.1 The μ-Metric

Definition 22.1 (μ-Distance (MuGeometry.v)). The Coq formalization defines `mu_distance` as the absolute difference of μ-totals:

$$d_\mu(S_1, S_2) = |\mu_{\text{total}}(S_2) - \mu_{\text{total}}(S_1)|$$

This is a one-dimensional projection, not a minimum over paths.

Theorem 22.1 (Pseudometric Properties (MuGeometry.v)). *d_μ satisfies non-negativity, reflexivity ($d(S, S) = 0$), symmetry, and triangle inequality—all inherited from absolute value on $\mathbb{Z}$. Note: d_μ is a **pseudometric**, not a metric: $d(S_1, S_2) = 0$ does* not *imply $S_1 = S_2$ (many distinct states share the same μ-total). The "geometry" is a 1D projection onto the μ counter.*

22.2 Causal Cones

Definition 22.2 (μ-Reachability Cone (SpacetimeEmergence.v)). $C^+(S) = \{S' \mid \exists \tau.\ S \xrightarrow{\tau} S' \wedge \text{cost}(\tau) \leq \text{Budget}\}$.

This is a structural analogy to causal cones: the set of reachable states given a finite μ-budget plays a role analogous to a light cone. The analogy should not be over-read—there is no Lorentz invariance, no metric signature, and no boost transformations in the Coq formalization.

22.3 No-Signaling

Theorem 22.2 (Observational Locality (SpacetimeEmergence.v)). *If module M_B is not in the target set of any instruction in a trace, then M_B's observable region is unchanged after execution (`exec_trace_no_signaling_outside_cone`).*

Remark 22.1. This is a **data isolation** property: operations targeting module A in the partition graph do not affect the lookup of module B. The proof is structural (list operations on disjoint keys), non-trivial mainly for `psplit`/`pmerge` which restructure the graph. The label "no-signaling" is an analogy—there is no spatial metric or speed limit between modules.

22.4 Four-Dimensional Spacetime Geometry

The previous sections treated spacetime as a 1D projection via the μ-metric. We now extend to full 4D geometry with discrete differential structure, demonstrating discrete differential geometry from computational primitives.

22.4.1 4D Simplicial Complex

Definition 22.3 (4D Simplicial Complex (FourDSimplicialComplex.v))**.** A **4D simplicial complex** $\mathcal{K}$ is a discrete approximation to 4D spacetime consisting of:

- **Vertices** V: Computational modules (ModuleID)
- **Edges** $E \subseteq V \times V$: Connections between modules
- **Simplices**: 2-simplices (triangles), 3-simplices (tetrahedra), 4-simplices (4D cells)

The complex satisfies:

- **Consistency**: Every face of a simplex is also in $\mathcal{K}$
- **Well-formedness**: Dimension 4 with locally Euclidean structure

22.4.2 Metric Tensor from μ-Costs

Definition 22.4 (Computational Metric (MetricFromMuCosts.v, RiemannTensor4D.v))**.** The metric tensor at vertex v with respect to vertices w_1, w_2 is:

$$g_{\mu\nu}(v; w_1, w_2) = \begin{cases} \ell(v, w_1, w_2) & \text{if } \mu = \nu \\ 0 & \text{if } \mu \neq \nu \end{cases} \tag{22.1}$$

where the edge length function is:

$$\ell(s, v_1, v_2) = 2 \cdot \text{mass}(v_1) \quad \text{if } v_1 = v_2 \tag{22.2}$$

and mass is derived from structural complexity: $\text{mass}(v) = |\text{module\_structural\_mass}(s, v)|$.

Remark 22.2. This metric is **position-dependent** for non-uniform mass distributions:

- **Uniform mass** m: $g_{\mu\mu}(v; w, w) = 2m$ (constant) $\implies$ flat spacetime
- **Non-uniform mass**: $g_{\mu\mu}(v; w, w)$ varies with $w \implies$ curved spacetime

This is the key mechanism by which computation creates geometry.

22.4.3 Discrete Differential Geometry

Definition 22.5 (Discrete Derivative (RiemannTensor4D.v))**.** For a scalar field $f : V \to \mathbb{R}$ and direction μ:

$$\partial_\mu f(v) = \frac{1}{|N_\mu(v)|} \sum_{w \in N_\mu(v)} (f(w) - f(v)) \tag{22.3}$$

where $N_\mu(v)$ is the set of neighbors of v in the μ-direction.

Definition 22.6 (Christoffel Symbols (RiemannTensor4D.v))**.** The connection coefficients (Christoffel symbols of the second kind) are:

$$\Gamma^\rho_{\mu\nu}(v) = \frac{1}{2} \sum_\sigma g^{\rho\sigma}(v) \Big(\partial_\mu g_{\nu\sigma}(v) + \partial_\nu g_{\mu\sigma}(v) - \partial_\sigma g_{\mu\nu}(v) \Big) \tag{22.4}$$

where $g^{\rho\sigma}$ is the inverse metric (diagonal for our construction).

Lemma 22.3 (Flat Spacetime Has Zero Connection (EinsteinEquations4D.v) — **(S)**)**.** *For uniform mass distribution:*

$$\forall w,\ mass(w) = m \quad \implies \quad \Gamma^\rho_{\mu\nu}(v) = 0 \tag{22.5}$$

Proof. Uniform mass $\implies$ position-independent metric $\implies \partial_\mu g_{\nu\sigma} = 0 \implies \Gamma^\rho_{\mu\nu} = 0$. □

22.4.4 Curvature Tensors

Definition 22.7 (Riemann Curvature Tensor (RiemannTensor4D.v))**.** The Riemann curvature tensor measures the failure of parallel transport:

$$\begin{aligned} R^{\rho}_{\sigma\mu\nu}(v) = \partial_\mu \Gamma^{\rho}_{\nu\sigma}(v) - \partial_\nu \Gamma^{\rho}_{\mu\sigma}(v) \\ + \sum_{\lambda} \Big(\Gamma^{\rho}_{\mu\lambda}(v)\Gamma^{\lambda}_{\nu\sigma}(v) - \Gamma^{\rho}_{\nu\lambda}(v)\Gamma^{\lambda}_{\mu\sigma}(v) \Big) \end{aligned} \tag{22.6}$$

Definition 22.8 (Ricci Tensor (RiemannTensor4D.v))**.** Contraction of the Riemann tensor:

$$R_{\mu\nu}(v) = \sum_{\rho} R^{\rho}_{\mu\rho\nu}(v) \tag{22.7}$$

Definition 22.9 (Ricci Scalar (RiemannTensor4D.v))**.** Complete contraction:

$$R(v) = \sum_{\mu,\nu} g^{\mu\nu}(v) R_{\mu\nu}(v) \tag{22.8}$$

Definition 22.10 (Einstein Tensor (RiemannTensor4D.v))**.** The Einstein tensor encodes spacetime curvature:

$$G_{\mu\nu}(v) = R_{\mu\nu}(v) - \frac{1}{2} g_{\mu\nu}(v) R(v) \tag{22.9}$$

Lemma 22.4 (Flat Spacetime Has Zero Curvature (EinsteinEquations4D.v) — **(S)**)**.** *For uniform mass:*

$$\forall w,\ \mathit{mass}(w) = m \quad \Longrightarrow \quad R^{\rho}_{\sigma\mu\nu}(v) = 0 \quad \Longrightarrow \quad G_{\mu\nu}(v) = 0 \tag{22.10}$$

22.4.5 Stress-Energy Tensor

Definition 22.11 (Energy Density (EinsteinEquations4D.v))**.** The energy density is the computational mass:

$$T_{00}(v) = \mathrm{mass}(v) \tag{22.11}$$

Definition 22.12 (Momentum Density (EinsteinEquations4D.v))**.** Spatial derivatives of energy:

$$T_{0i}(v) = T_{i0}(v) = \partial_i T_{00}(v) \tag{22.12}$$

Definition 22.13 (Stress Components (EinsteinEquations4D.v))**.** Spatial stress tensor:

$$T_{ij}(v) = \begin{cases} \mathrm{mass}(v) & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \tag{22.13}$$

Definition 22.14 (Stress-Energy Tensor (EinsteinEquations4D.v))**.** Combining all components:

$$T_{\mu\nu}(s, \mathcal{K}, v) = \begin{cases} T_{00}(v) & \text{if } \mu = \nu = 0 \\ T_{0i}(v) & \text{if } \mu = 0, \nu = i \text{ or } \mu = i, \nu = 0 \\ T_{ij}(v) & \text{if } \mu = i, \nu = j \\ 0 & \text{otherwise} \end{cases} \tag{22.14}$$

22.4.6 Einstein's Field Equations

Definition 22.15 (Gravitational Constant (EinsteinEquations4D.v))**.** In computational units where the fundamental scale is set by μ-costs:

$$G = \frac{1}{8\pi} \tag{22.15}$$

This makes $8\pi G = 1$, normalizing the coupling in Einstein's equations.

Lemma 22.5 (Curvature from μ-Gradients (EinsteinEquations4D.v) — **(S)**)**.** *For uniform mass distribution:*

$$\forall w,\ \mathit{mass}(w) = m \implies G_{\mu\nu}(v) = 0 \tag{22.16}$$

Proof. The proof chain is:

1. Uniform mass $\implies$ position-independent metric
2. Position-independent metric $\implies$ zero discrete derivatives $\partial_\mu g_{\nu\sigma} = 0$
3. Zero metric derivatives $\implies$ zero Christoffel symbols $\Gamma^\rho_{\mu\nu} = 0$
4. Zero Christoffel $\implies$ zero Riemann tensor $R^\rho_{\sigma\mu\nu} = 0$
5. Zero Riemann $\implies$ zero Ricci tensor and scalar
6. Zero Ricci $\implies$ zero Einstein tensor $G_{\mu\nu} = 0$

All steps are proven constructively in `EinsteinEquations4D.v`. □

Lemma 22.6 (Stress-Energy Conservation (EinsteinEquations4D.v) — **(S)**)**.** *For vacuum states (zero mass everywhere):*

$$\forall w,\ \mathit{mass}(w) = 0 \implies T_{\mu\nu}(v) = 0 \tag{22.17}$$

Proof. Direct computation:

- $T_{00} = \text{mass} = 0$
- $T_{0i} = \partial_i(\text{mass}) = \partial_i(0) = 0$
- $T_{ij} = \delta_{ij} \cdot \text{mass} = \delta_{ij} \cdot 0 = 0$

□

Theorem 22.7 (Einstein Equations in Vacuum (EinsteinEquations4D.v) — **(S)**)**.** *For any VM state s, 4D simplicial complex $\mathcal{K}$, spacetime indices μ, ν, and vertex v:*

$$\boxed{\forall w,\ \mathit{mass}(w) = 0 \implies G_{\mu\nu}(s, \mathcal{K}, v) = 8\pi G\, T_{\mu\nu}(s, \mathcal{K}, v)} \tag{22.18}$$

Note: *This is the vacuum case where both sides equal zero ($0 = 0$). The gravitational constant $G := 1/(8\pi)$ is a* ***unit choice*** *(a definition), not a derived quantity. In vacuum, all masses vanish, so $T_{\mu\nu} = 0$ and uniform-mass (zero) yields $G_{\mu\nu} = 0$; the equation reduces to $0 = 8\pi \cdot \frac{1}{8\pi} \cdot 0 = 0$. The non-vacuum case is* ***not proven*** *and would require discrete Bianchi identities not present in the formalization.*

Proof. **Step 1**: Expand the right-hand side using $G = 1/(8\pi)$:

$$8\pi G\, T_{\mu\nu} = 8\pi \cdot \frac{1}{8\pi} \cdot T_{\mu\nu} = T_{\mu\nu} \tag{22.19}$$

Step 2: Apply Lemma 22.5 (curvature from μ-gradients):
Vacuum (mass = 0 everywhere) is a special case of uniform mass distribution, so:

$$G_{\mu\nu}(v) = 0 \tag{22.20}$$

Step 3: Apply Lemma 22.6 (stress-energy conservation):
For vacuum:

$$T_{\mu\nu}(v) = 0 \tag{22.21}$$

Conclusion: Both sides equal zero:

$$0 = 0 \quad \checkmark \tag{22.22}$$

The complete proof is formalized in `coq/kernel/EinsteinEquations4D.v` with:

- No project-local axioms
- Zero admitted lemmas
- Zero classical logic imports
- Verified by automated proof auditing (Inquisitor: 0 HIGH findings)

□

Remark 22.3 (Vacuum vs. Non-Vacuum)**.** Theorem 22.7 requires the vacuum hypothesis (mass = 0 everywhere). For non-vacuum states with arbitrary mass distributions, the equation $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ only holds when the distribution satisfies discrete Bianchi identities (conservation laws). This is analogous to classical general relativity, where Einstein's equations are *field equations* that constrain which configurations are physically possible, not universal identities satisfied by all states.

The vacuum case is sufficient to demonstrate that spacetime curvature emerges from computational dynamics. The non-vacuum case requires discrete differential geometry beyond the current formalization (specifically, discrete covariant derivatives and discrete Bianchi identities).

Remark 22.4 (Epistemological Status)**.** This is a **(S) Structural** result: it proves a mathematical relationship between computational primitives (VM states, μ-costs) and geometric quantities (curvature tensors) with no physical axioms. The theorem establishes that:

1. Spacetime geometry (metric, curvature) can be constructed from pure computation
2. The resulting geometry satisfies the vacuum Einstein equation in the formally verified zero-mass regime
3. Beyond that vacuum regime, the repository contains a conditional full-tensor bridge with named premises, a proved uniform diagonal boundary witness in the active operator, and a separately proved non-uniform boundary witness in a concrete affine metric-scaled redesign branch rather than one universal unconditional theorem

The result does *not* claim that physical spacetime *is* implemented by a Turing machine—that would be a **(C)** conditional claim requiring empirical validation. More importantly, it does not justify the stronger statement that arbitrary computational substrates with μ-cost tracking automatically satisfy the full Einstein equations in complete generality. The honest status is narrower: the vacuum theorem is unconditional; the full-tensor bridge in `EinsteinEquationsFull.v` remains conditional; the active first-neighbor operator proves the uniform diagonal boundary witness and formally refutes the stronger non-uniform diagonal extension; the broader non-negative local weighted family containing the simple symmetric average also fails on the same shell; and `SymmetricDerivative4D.v` proves that a concrete signed affine metric-scaled redesign rescues that non-uniform boundary witness with off-diagonal Ricci vanishing, Einstein tensor $G_{\mu\nu} = 3\,\delta_{\mu\nu}$ at the critical vertex, and, under `module_structural_mass s 1 = 1`, the aligned full-tensor witness $G_{\mu\nu} = 3\,T_{\mu\nu}$ there.

Corollary 22.8 (Discrete Vacuum Geometry is Emergent)**.** *Discrete vacuum spacetime geometry is an emergent property of the machine's information-processing structure.*

Proof. Theorem 22.7 constructs the discrete metric, curvature, and vacuum Einstein relation from computational primitives (VM states, operations, μ-costs) with no physical axioms. Therefore, the repo's discrete vacuum geometry is a derived concept rather than a primitive one. □

22.4.7 Empirical Validation

The theoretical results are validated numerically:

Theorem 22.9 (Numerical Verification (tests/test_*) — Empirical)**.**

1. ***Flat spacetime*** *(uniform mass $m = 1$):*
   - $\max|\Gamma^\rho_{\mu\nu}| < 10^{-10}$ ✓
   - $\max|R^\rho_{\sigma\mu\nu}| < 10^{-10}$ ✓
   - $\max|G_{\mu\nu}| < 10^{-10}$ ✓
2. ***Curved spacetime*** *(non-uniform mass $m \in [0.5, 1.5]$):*
   - $\max|\Gamma^\rho_{\mu\nu}| \approx 3.0$ ✓ *(curvature detected)*
   - $\max|R^\rho_{\sigma\mu\nu}| > 0$ ✓
3. ***Vacuum Einstein equations*** *(mass $= 0$):*
   - $|G_{\mu\nu}| < 10^{-10}$ ✓
   - $|T_{\mu\nu}| < 10^{-10}$ ✓
   - $|G_{\mu\nu} - 8\pi G T_{\mu\nu}| < 10^{-10}$ ✓

These tests confirm that the discrete differential geometry correctly implements the theoretical predictions.

Chapter 23

Information Causality

Theorem 23.1 (IC–μ Equivalence (InformationCausality.v) — **(R)**). *The file defines two record types (`ICScenario` and `MuScenario`) with opaque `Prop` fields, then proves they are equivalent* given *a hypothesis (`ic_mu_equivalent`) that literally states the equivalence. The proof is `tauto`. No mutual information, probability distributions, or CHSH values appear anywhere in the formalization.*

Remark 23.1. The intended statement—that Bob's total information about Alice's data is bounded by the channel's μ-capacity—is a reasonable physical claim but is **not formalized** in the Coq file. The current file is a placeholder that assumes the conclusion as a hypothesis and re-derives it. A future version would need to define entropy, mutual information, and prove the bound from the VM semantics.

Chapter 24

Thermodynamic Bridge

24.1 Landauer's Principle

Theorem 24.1 (Landauer Derived (LandauerDerived.v))**.**

1. `erasure_irreversible`*: Erasing* ≥ 1 *bit implies fan-in* $= 2^k > 1$ *(arithmetic:* $2^k > 1$ *for* $k \geq 1$*).*

2. `erasure_decreases_entropy`*: Erasure decreases system entropy (*$\Delta S < 0$*) — this is just* $\text{output} < \text{input} \Rightarrow \text{output} - \text{input} < 0$.

3. `landauer_information_bound`*: For any physical erasure, environment entropy increase* $\geq$ *bits erased.*

4. `erasure_additive`*: Sequential erasures compose additively.*

Caveat on item 3*: The* `PhysicalErasure` *record has* `second_law_satisfied` *as a* ***field****, not a derived property. The theorem* `landauer_information_bound` *simply extracts this field, making the bound an* input assumption *rather than a derivation. The physical energy bound* $Q \geq k_B T \ln 2 \cdot n$ *is* ***explicitly not proven*** *(the file itself notes the conversion factor is not formalized). Also, the* `mu` *in this file is a standalone synonym for* `bits_erased`*—it has no formal connection to* `vm_mu` *in* `VMState.v`*.*

24.2 Dissipative Embedding

Theorem 24.2 (Dissipative Model (DissipativeEmbedding.v — reference-only))**.** *Irreversible (dissipative) physics embeds into the Thiele VM with* $\Delta\mu \geq 1$ *per irreversible step. Reversible physics embeds with* $\Delta\mu = 0$*.*

Theorem 24.3 (Reversible Physics (PhysicsEmbedding.v — reference-only, WaveEmbedding.v — reference-only))**.** *Reversible lattice gas and wave models embed with* $\mu_{\text{final}} = \mu_{\text{init}}$ *(zero cost). Particle count, momentum, wave energy, and wave momentum are conserved by the VM.*

Chapter 25

Self-Reference and the Thiele Manifold

25.1 Self-Reference Escalation

Theorem 25.1 (Gödel-Style Escalation (SelfReference.v)). *Any self-referential system S requires a meta-system* Meta *with strictly more dimensions:*

$$\text{self\_ref}(S) \implies \exists\, \text{Meta} : \dim(\text{Meta}) > \dim(S)$$

The meta-system inherits self-reference, creating an infinite escalation.

25.2 The Thiele Manifold

Definition 25.1 (Thiele Manifold (ThieleManifold.v)). An infinite tower of systems $\{L_n\}_{n\in\mathbb{N}}$ with:

- $\dim(L_n) = 4 + n$ (level n has dimension $4 + n$).
- $\dim(L_{n+1}) > \dim(L_n)$ (strict enrichment).
- Each level reasons about the level below.
- Self-reference at level n is answered by level $n + 1$.

Theorem 25.2 (Projection to Spacetime (ThieleManifold.v)). *The projection π_4 collapses the tower to dimension 4 (spacetime). For $n > 0$:*

- *π_4 is lossy:* $\dim(L_n) > 4$.
- *μ-cost of projection* $= \dim(L_n) - 4 = n > 0$.

Spacetime is the "shadow" of the full manifold.

25.3 Genesis: Process–Machine Isomorphism

Theorem 25.3 (Genesis (Genesis.v — reference-only)). *There is a definitional isomorphism* Proc $\cong$ Thiele *between coherent processes (step + admissibility proof) and Thiele machines:*

$$\text{thiele\_to\_proc}(\text{proc\_to\_thiele}(P)) = P \tag{25.1}$$

$$\text{proc\_to\_thiele}(\text{thiele\_to\_proc}(T, H)) = T \tag{25.2}$$

Any coherent process is *a Thiele machine, and vice versa.*

Chapter 26

Three-Layer Isomorphism

Theorem 26.1 (Observable Comparison Contract (ThreeLayerIsomorphism.v))**.** *The active Coq theorem packages the* observable-state contract *used for cross-layer comparison: if the chosen Python and RTL decoders satisfy the same projection contract as the Coq layer, then the decoded observable states agree on that projection for a shared trace τ.*

$$\mathrm{decode}_{\mathrm{Python}}(\tau) = \mathrm{decode}_{\mathrm{Coq}}(\tau) \wedge \mathrm{decode}_{\mathrm{RTL}}(\tau) = \mathrm{decode}_{\mathrm{Coq}}(\tau) \Longrightarrow \mathrm{decode}_{\mathrm{Python}}(\tau) = \mathrm{decode}_{\mathrm{RTL}}(\tau)$$

This is a contract theorem over explicit observable projections, not an unconditional claim that every current Python and RTL artifact is universally equal to Coq in all CI environments.

Chapter 27

The Curry–Howard–Thiele Correspondence

Theorem 27.1 (Logic–Computation Isomorphism (LogicIsomorphism.v — reference-only))**.** *The Thiele Machine extends the Curry–Howard correspondence:*

Logic	***Computation***	***Thiele***
Proposition	*Type*	*Partition*
Proof	*Program (term)*	*Trace*
Cut elimination	*β-reduction*	*Execution (*Run*)*
Proof equivalence	$=_\beta$	*Execution equivalence*

A valid proof corresponds to a terminating execution. Proof equivalence $\Leftrightarrow$ execution equivalence.

Theorem 27.2 (Logic Is Physics (LogicToPhysics.v — reference-only))**.** *Cut elimination in logic = relational composition in physics:*

$$\mathrm{interp}(\mathrm{cut}(\pi_1, \pi_2)) = \mathrm{rel\_comp}(\mathrm{interp}(\pi_1), \mathrm{interp}(\pi_2))$$

Logical proof composition maps to physical relation composition. This is the formal nucleus of the claim that logic and physics share the same categorical structure.

Chapter 28

Categorical Structure

28.1 The Morphism Layer

The active implementation adds a concrete category directly to the machine state. This is not an abstract analogy — it is an executable layer of 7 opcodes (0x27–0x2D) with formally verified laws.

28.1.1 Objects and Morphisms

The partition graph G now carries two maps:

- `pg_modules` : ModuleID $\rightharpoonup$ ModuleState — objects (partition modules)
- `pg_morphisms` : MorphismID $\rightharpoonup$ MorphismState — arrows

A MorphismState bundles: source module, target module, coupling data (list of ($nat \times nat$) pairs plus a label), and an identity flag.

28.1.2 The Seven Categorical Opcodes

- `MORPH`($dst, src, tgt, idx, \Delta\mu$): create morphism $src \to tgt$, write morphism ID to `regs`[dst]. Costs $\Delta\mu$.
- `COMPOSE`($dst, f, g, \Delta\mu$): compose morphisms $f; g$ when f.target = g.source, write new ID to `regs`[dst]. Costs $\Delta\mu$. Errors if endpoints mismatch.
- `MORPH_ID`($dst, m, \Delta\mu$): create identity morphism id_m for module m (diagonal relation). Costs $\Delta\mu$.
- `MORPH_DELETE`($f, \Delta\mu$): remove morphism f. Cascades on module deletion via `graph_cascade_delete_mor`
- `MORPH_ASSERT`($f, prop, cert, \Delta\mu$): cert-setter. Costs $S(\Delta\mu) \geq 1$ unconditionally — even for failed attempts (morphism does not exist). This is the No Free Insight law applied to relational claims.
- `MORPH_TENSOR`($dst, f, g, \Delta\mu$): tensor product $f \otimes g$. Requires that the union modules $\mathrm{src}(f) \cup \mathrm{src}(g)$ and $\mathrm{tgt}(f) \cup \mathrm{tgt}(g)$ already exist in the graph.
- `MORPH_GET`($dst, f, sel, \Delta\mu$): read field from morphism f: selector 0 = source, 1 = target, 2 = coupling length, 3 = is_identity.

28.1.3 Category Laws

Theorem 28.1 (Relational Composition is Associative (kernel/CategoryLaws.v) — **(S)**)**.** *For any three morphisms $f : A \to B$, $g : B \to C$, $h : C \to D$ with composable endpoints:*

$$(f; g); h = f; (g; h)$$

in the relational sense (coupling pairs of the composite). Zero Admitted.

Theorem 28.2 (Graph Composition Satisfies Laws (kernel/CategoryBridge.v) — **(S)**)**.** *The operation `graph_compose_morphisms` satisfies associativity and unitality. Additionally:*

`categorical_extension_nofi_consistent`

Any chain of MORPH + COMPOSE operations yields $\mu \geq$ number of operations — the NoFI policy applies to morphism construction.

Theorem 28.3 (Tensor Bifunctoriality (kernel/CategoryMonoidal.v) — **(S)**)**.** `graph_tensor_morphisms` *is a bifunctor: it satisfies the interchange law*

$$(f \otimes g); (h \otimes k) = (f; h) \otimes (g; k)$$

for composable, disjoint pairs.

28.1.4 The Categorical Separation Theorem

Theorem 28.4 (Categorical Separation (kernel/PartitionSeparation.v, §10) — **(S)**)**.** *There exist states s_1, s_2 such that:*

$$s_1.\texttt{vm\_regs} = s_2.\texttt{vm\_regs} \wedge s_1.\texttt{vm\_mem} = s_2.\texttt{vm\_mem} \wedge s_1.\texttt{vm\_mu} = s_2.\texttt{vm\_mu} \wedge s_1.\texttt{vm\_err} = s_2.\texttt{vm\_err}$$

but $s_1.\texttt{vm\_graph.pg\_morphisms} \neq s_2.\texttt{vm\_graph.pg\_morphisms}$.

Proof. Construct s_1 with `pg_morphisms` $= [(0, ms)]$ for any morphism state ms, and s_2 with `pg_morphisms` $= []$. Arrange identical `vm_regs`, `vm_mem`, `vm_mu`, `vm_err` by choosing costs and instruction sequences accordingly. Distinction by `discriminate`. □ □

Remark 28.1 (Significance)**.** A classical machine (exposing only registers, memory, and a cost counter) cannot distinguish s_1 from s_2. The Thiele Machine distinguishes them in one probe instruction: `MORPH_DELETE` 0 succeeds on s_1 and errors on s_2. The categorical layer records *how* knowledge was derived, not just *what* the final values are.

This is the machine's claim to being a *knowledge auditor*: any party claiming to have verified a structural relation must carry a μ-receipt $\geq$ the minimum construction cost. The receipt is unforgeable by the NoFI theorem applied to `MORPH_ASSERT`.

28.2 The Conservation as Functor (Archived)

Theorem 28.5 (Conservation as Functor (Universe.v — reference-only))**.** *Define two categories:*

- $\mathbf{C}_{\text{phys}}$*: Objects = universe states (particle momenta lists). Morphisms = paths of momentum-conserving interactions.*
- $\mathbf{C}_{\text{logic}}$*: Objects = total momentum values. Morphisms = equality proofs.*

The functor $F : \mathbf{C}_{\text{phys}} \to \mathbf{C}_{\text{logic}}$ maps $F(s) = \sum s$ (total momentum). Then:

$$\text{Path}(s_1, s_2) \implies \sum s_1 = \sum s_2$$

Conservation of momentum emerges as the functorial image. Observation is a structure-preserving map from physics to logic.

Chapter 29

Audit Infrastructure

Theorem 29.1 (CatNet Integrity (CatNet.v — reference-only))**.** *Adding a new entry to a valid hash-chain audit log produces a valid chain. If the logic oracle detects inconsistency, execution halts and no further entries are written (paradox halting).*

Chapter 30

Falsifiable Predictions

30.1 Linear Scaling of Structural Cost

Axiom 30.1 (Linear Scaling (kernel/FalsifiablePrediction.v)). The μ-cost of maintaining coherence of N entangled qubits scales as $O(N)$ per time step.

Prediction: Large-scale quantum computers will encounter a fundamental (not merely technical) decoherence noise floor proportional to entanglement complexity.

30.2 CHSH Regime Separation

Prediction: Experiments probing the boundary between quantum ($|S| \leq 2\sqrt{2}$) and supra-quantum ($|S| > 2\sqrt{2}$) correlations should find that exceeding the Tsirelson bound requires measurably higher energy dissipation, scaling as:

$$\Delta E \geq k_B T \ln 2 \cdot (|S| - 2\sqrt{2})$$

30.3 Metric Deformation

Since d_μ depends on information content, regions of high structural complexity effectively expand the metric.

Prediction: In high-complexity computations, effective signal latency will increase relative to vacuum propagation. (This prediction relies on interpreting d_μ as physically meaningful, which is currently an analogy rather than a derived result.)

30.4 Architectural Permanence

Theorem 30.1 (Optimal Quartet (ArchTheorem.v — reference-only)). *The four partition discovery strategies (Louvain, Spectral, Degree, Balanced) achieve classification accuracy $> 90\%$ across all problem classes. No alternative configuration exceeds this quartet's accuracy. The configuration is architecturally final.*

30.5 Coupling Constant Prediction

Definition 30.1 (Thiele α Limit (thiele_manifold/PhysicalConstants.v)). The asymptotic density of self-referential programs in the n-bit state space:

$$\alpha = \lim_{n\to\infty} \frac{A_{\text{interaction}}(n)}{V_{\text{spacetime}}(n)} = \lim_{n\to\infty} \frac{n+1}{2^n}$$

This converges to 0, but the *non-asymptotic* structure at physical scales should relate to coupling constants.

Appendix A

Proof File Index

The table below maps each theorem to its Coq source file. Entries marked — *reference-only* are theoretical results that belong to an extended corpus; the corresponding Coq files are not present in the active kernel tree and their proofs are not machine-checked in the current build. Everything unmarked lives in the active corpus and compiles clean.

Result	**Coq Source**
No Free Insight (Thm. 5.1)	`nofi/NoFreeInsight_Theorem.v`
μ-Chaitin	`nofi/MuChaitinTheory_Theorem.v`
Cost = Complexity	`theory/CostIsComplexity.v`
No Free Lunch	`theory/NoFreeLunch.v`
No Arbitrage	`kernel/NoArbitrage.v`
TM Embedding	`thielemachine/coqproofs/Embedding_TM.v`
TM $\rightarrow$ Minsky	`modular_proofs/TM_to_Minsky.v`
Strict Subsumption	`thielemachine/coqproofs/Subsumption.v`
Semantic Strictness	`thielemachine/coqproofs/ThieleFoundations.v`
Confluence	`thielemachine/coqproofs/Confluence.v`
Tensoriality	`thielemachine/coqproofs/ThieleUnificationTensor.v`
μ-Conservation	`kernel/MuLedgerConservation.v`
Receipt Integrity	`kernel/ReceiptIntegrity.v`
Noether / Gauge	`kernel/KernelNoether.v`
Gauge Trace Preservation	`thielemachine/coqproofs/RepresentationTheorem.v`
2D Necessity	`quantum_derivation/TwoDimensionalNecessity.v`
Complex Necessity	`quantum_derivation/ComplexNecessity.v`
Born Rule	`kernel/BornRule.v`
Born Rule (Tensor)	`kernel/BornRuleFromSymmetry.v`
Unitarity	`kernel/Unitarity.v`
Purification	`kernel/Purification.v`
No-Cloning	`kernel/NoCloning.v`
No-Cloning (Machine-Native)	`kernel/NoCloningFromMuMonotonicity.v`
Observation Irreversibility	`quantum_derivation/ObservationIrreversibility.v`
Collapse Determination	`quantum_derivation/CollapseDetermination.v`
Tsirelson Derivation	`kernel/TsirelsonDerivation.v`
Tsirelson (Algebraic)	`kernel/TsirelsonFromAlgebra.v`
Tsirelson (General)	`kernel/TsirelsonGeneral.v`
CHSH Separation	`thielemachine/verification/Deliverable_CHSHSeparation.v`
Emergent Schrödinger	`physics_exploration/EmergentSchrodinger.v`
Planck Consistency	`physics_exploration/PlanckDerivation.v`
Spacetime Emergence	`kernel/SpacetimeEmergence.v`
μ-Geometry	`kernel/MuGeometry.v`
4D Simplicial Complex	`kernel/FourDSimplicialComplex.v`
Metric from μ-Costs	`kernel/MetricFromMuCosts.v`
Riemann Curvature Tensor	`kernel/RiemannTensor4D.v`
Christoffel Symbols	`kernel/RiemannTensor4D.v`
Einstein Tensor	`kernel/RiemannTensor4D.v`
Stress-Energy Tensor	`kernel/EinsteinEquations4D.v`
Curvature from μ-Gradients	`kernel/EinsteinEquations4D.v`
Stress-Energy Conservation	`kernel/EinsteinEquations4D.v`
Einstein Equations (Vacuum)	`kernel/EinsteinEquations4D.v`
Einstein Equations (General)	`kernel/EinsteinEquationsFull.v`
Discrete Gaussian Curvature	`kernel/MetricFromMuCosts.v`
Information Causality	`kernel/InformationCausality.v`
Landauer	`thermodynamic/LandauerDerived.v`
Dissipative Embedding	`thielemachine/coqproofs/DissipativeEmbedding.v`
Physics Embedding	`thielemachine/coqproofs/PhysicsEmbedding.v`
Wave Embedding	`thielemachine/coqproofs/WaveEmbedding.v`
Self-Reference	`self_reference/SelfReference.v`
Thiele Manifold	`thiele_manifold/ThieleManifold.v`
Genesis	`theory/Genesis.v`
Full Isomorphism	`kernel/ThreeLayerIsomorphism.v`
Curry–Howard–Thiele	`thielemachine/coqproofs/LogicIsomorphism.v`
Logic Is Physics	`theory/LogicToPhysics.v`
Functor Soundness	`isomorphism/coqproofs/Universe.v`
CatNet Integrity	`catnet/coqproofs/CatNet.v`
Amortization	`thielemachine/coqproofs/AmortizedAnalysis.v`
Optimal Quartet	`theory/ArchTheorem.v`
Falsifiable Prediction	`kernel/FalsifiablePrediction.v`
Physical Constants	`thiele_manifold/PhysicalConstants.v`
Halting Undecidability	`kernel/OracleImpossibility.v`
Oracle Cost Bound	`kernel/OracleImpossibility.v`
Hyper-Thiele Oracle	`thielemachine/coqproofs/HyperThiele_Halting.v`

Appendix B

Epistemological Classification

The following table classifies every major result by its epistemological status.

Result	Status	Key Assumption
Unconditional structural theorems about the machine		
No Free Insight	**(S)**	Module type contract
μ-Chaitin	**(S)**	Module type contract
Cost = Complexity	**(S)**	Prefix-free coding
No Free Lunch	**(S)**	Faithful representation
No Arbitrage ⇒ Potential	**(S)**	Additivity + no-arbitrage (toy model)
TM Embedding	**(S)**	Standard simulation
Strict Subsumption	**(S)**	Partition structure
Halting Undecidability	**(S)**	Diagonalization
Oracle Cost	**(S)**	Definitional (soundness := cost ≥ 1)
Confluence	**(S)**	Module independence
μ-Conservation	**(S)**	Machine semantics only
Receipt Integrity	**(S)**	Hash-chain model
Gauge / Noether	**(S)**	Definitional (record structure)
Genesis	**(S)**	Definitional isomorphism
Three-Layer Isomorphism / comparison contract	**(C)**	Backend decoders satisfy the shared observable projectio
Curry–Howard–Thiele	**(S)**	Interpretation map
Curvature from μ-Gradients	**(S)**	Lemma 22.5
Stress-Energy Conservation	**(S)**	Lemma 22.6
Einstein Equations (Vacuum)	**(S)**	Theorem 22.7
Einstein Equations (General full tensor)	**(C)**	Named off-diagonal Ricci premise in `EinsteinEquations`
Discrete Differential Geometry	**(S)**	Definitions only
Conditional derivations (valid given stated axioms)		
2D Necessity	**(C)**	Superposition axiom (Axiom 12.1)
Complex Necessity	**(C)**	2D + norm preservation
Born Rule Uniqueness	**(C)**	Tensor product consistency (`BornRuleFromSymmetry.v`)
Unitarity from $\mu = 0$	**(C)**	Info conservation (one direction only)
No-Cloning	**(C)**	Purity = information
Purification	**(C)**	Bloch sphere model
Observation Irreversibility	**(C)**	REVEAL semantics
Collapse Determination	**(C)**	Entropy = $\log \dim$
Emergent Schrödinger	**(C)**	2D + antisymmetry
Landauer	**(C)**	Second law
emphassumed as record field		
Dissipative Embedding	**(C)**	Irreversibility model
Self-Reference	**(C)**	Dimension = complexity
CHSH Separation	**(C)**	Numerical witnesses
Tsirelson Bound	**(C)**	Algebraic derivation (`TsirelsonFromAlgebra.v`)
Consistency relations (definitions verified, not predictions)		
Planck's Constant	**(R)**	$\tau_\mu := h/(4E_L)$
Information Causality	**(R)**	Circular: equivalence assumed as hypothesis
μ-Geometry	**(R)**	Pseudometric (absolute value on $\mathbb{Z}$)
Spacetime Locality	**(R)**	Data isolation in partition graph
Born Rule (`BornRule.v`)	**(R)**	μ-consistent hypothesis unused

Legend: **(S)** = unconditional theorem about the machine; **(C)** = valid derivation conditional on stated axiom; **(R)** = internal consistency check.